

CVP Project

Complete Command Reference

All commands, shortcuts, and URLs used across the backend, frontend, security auditing, VS Code task runner, and monorepo setup.

Backend — Python / FastAPI / UV

Command / Shortcut	What it does
<code>uv sync</code>	Installs all Python dependencies from pyproject.toml into the virtual environment. Run after pulling new backend changes.
<code>.venv\Scripts\activate.bat</code>	Activates the Python virtual environment on Windows. Required before running backend commands manually.
<code>uv run fastapi dev</code>	Starts FastAPI dev server with hot reload. Backend runs at <code>http://localhost:8000</code> .
<code>uv run fastapi dev --port 8001</code>	Starts backend on a custom port if 8000 is already in use.
<code>uv add</code>	Adds a new Python package and updates pyproject.toml automatically.
<code>uv remove</code>	Removes a Python package from the project.
<code>uv run alembic upgrade head</code>	Runs all pending database migrations. Run after pulling changes that include new models.
<code>uv run alembic revision --autogenerate -m "msg"</code>	Creates a new migration file based on model changes.

Frontend — React / Vite / npm

Command / Shortcut	What it does
<code>npm install</code>	Installs all frontend dependencies from package.json. Run once after cloning or pulling changes.
<code>npm run dev</code>	Starts Vite dev server with hot reload. Frontend runs at <code>http://localhost:3000</code> .
<code>npm run build</code>	Compiles and bundles frontend for production. Output goes to <code>dist/</code> folder.
<code>npm run preview</code>	Serves the production build locally to test before deploying. Runs at <code>http://localhost:4173</code> .

Command / Shortcut	What it does
<code>npm run lint</code>	Runs Biome linting and formatting checks across all frontend files.
<code>npm run generate-client</code>	Generates TypeScript API client from backend OpenAPI schema. Run whenever backend routes or models change. Requires backend running.
<code>npm install</code>	Adds a new runtime dependency to the frontend project.
<code>npm install --save-dev</code>	Adds a new development-only dependency (build tools, type packages, etc.).
<code>npm uninstall</code>	Removes a package from the project.

Security — npm audit

Command / Shortcut	What it does
<code>npm audit</code>	Scans all installed packages for known vulnerabilities. Shows severity and affected packages.
<code>npm audit fix</code>	Automatically fixes vulnerabilities that don't require breaking changes. Safe to run anytime.
<code>npm audit fix --force</code>	Fixes all vulnerabilities including major version upgrades. May break things — test afterward.
<code>npm list</code>	Shows the installed version of a specific package and its position in the dependency tree.

VS Code — keyboard shortcuts

Command / Shortcut	What it does
<code>Ctrl + Shift + B</code>	Runs the default build task — starts both backend and frontend and opens the browser.
<code>Ctrl + Shift + P</code>	Opens the Command Palette — main way to access all VS Code commands.
<code>Ctrl + Shift + P → Tasks: Run Task</code>	Shows all available tasks — Start Backend, Start Frontend, Generate API Client, Restart tasks.
<code>Ctrl + Shift + G</code>	Custom shortcut to run Generate API Client instantly (if configured in <code>keybindings.json</code>).
<code>F5</code>	Opens Run and Debug panel — launch frontend or API docs in Chrome via <code>launch.json</code> .

Command / Shortcut	What it does
Ctrl + `	Toggles the integrated terminal. Use tabs to switch between Backend and Frontend output.
Ctrl + C (in terminal)	Stops the currently running process in that terminal tab.
Ctrl + Shift + X	Opens Extensions panel to install or manage VS Code extensions.

Root monorepo — package.json scripts

Command / Shortcut	What it does
npm run dev (from root)	Starts both backend and frontend together using concurrently. Color-coded output in a single terminal.
npm run dev:backend	Starts only the backend server from the root folder.
npm run dev:frontend	Starts only the frontend server from the root folder.
npm run generate-client (from root)	Runs API client generator from root level — no need to cd into the frontend folder.

Important URLs

Command / Shortcut	What it does
http://localhost:3000	Frontend React application — your main development URL.
http://localhost:8000/docs	FastAPI Swagger UI — interactive API docs. Test all endpoints directly in the browser.
http://localhost:8000/redoc	ReDoc API documentation — cleaner read-only view of all endpoints and schemas.
http://localhost:8000/openapi.json	Raw OpenAPI schema JSON — what the client generator reads to produce src/client/ files.

Run generate-client whenever backend routes or models change. Keep .env files in both cvp_backend/ and cvp_frontend/ for correct configuration.